

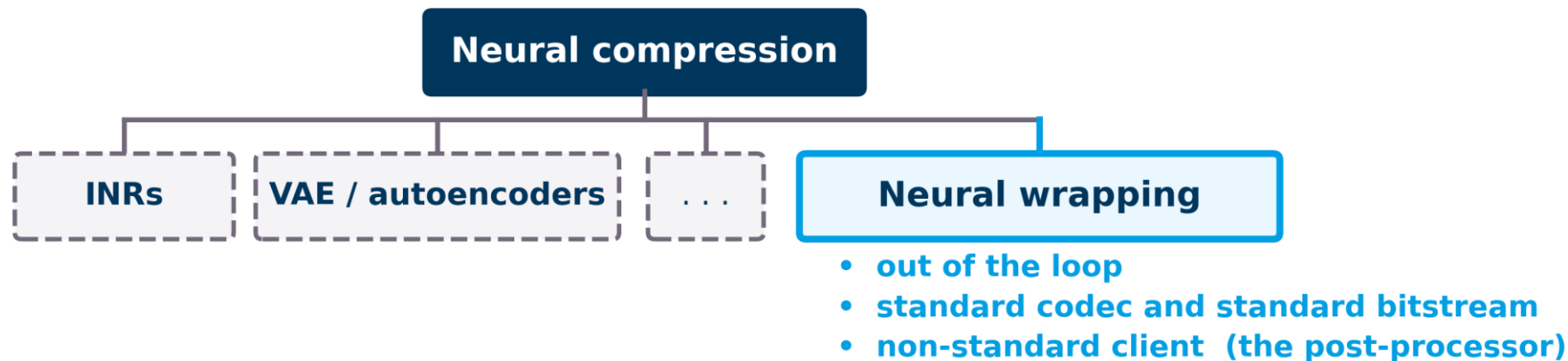
A Projection-Based Surrogate Gradient Interpretation for Neural Codec Wrappers

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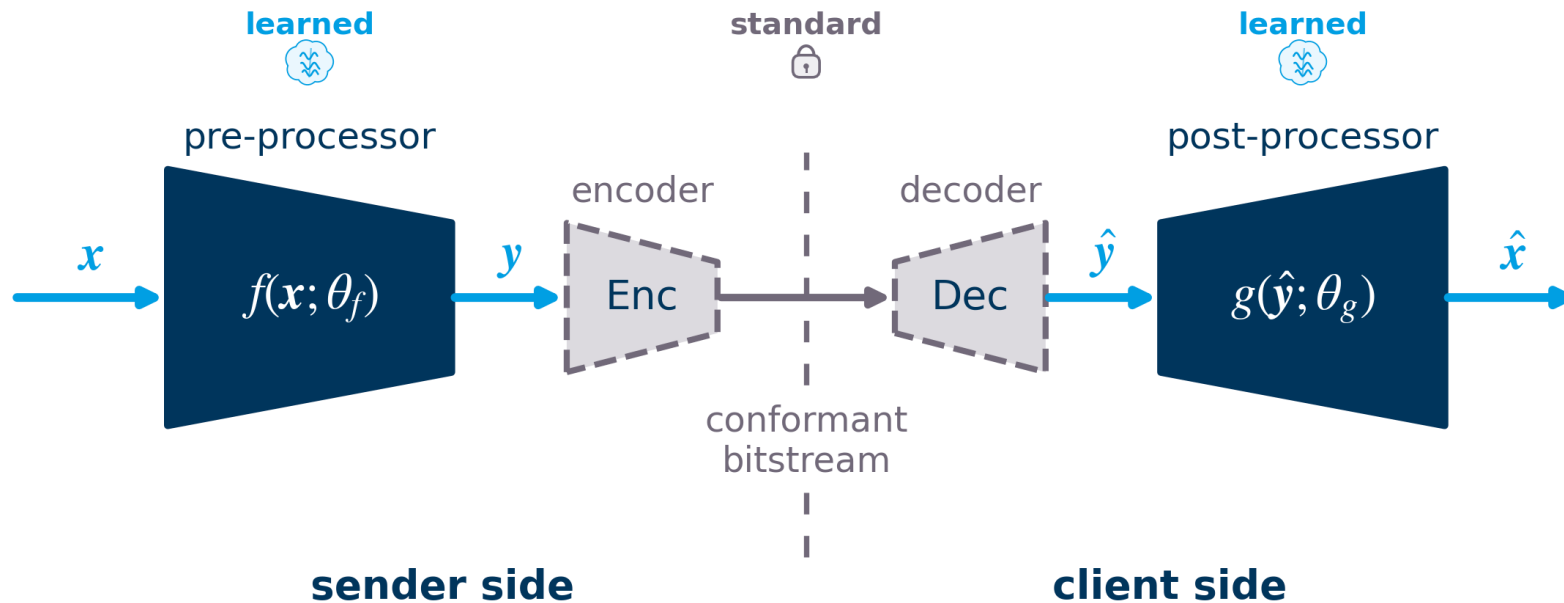
Neural wrapping: a branch of neural compression



Goal: improve the rate-distortion performance of an already-deployed encoder.

The gain has to come without changing the bitstream syntax

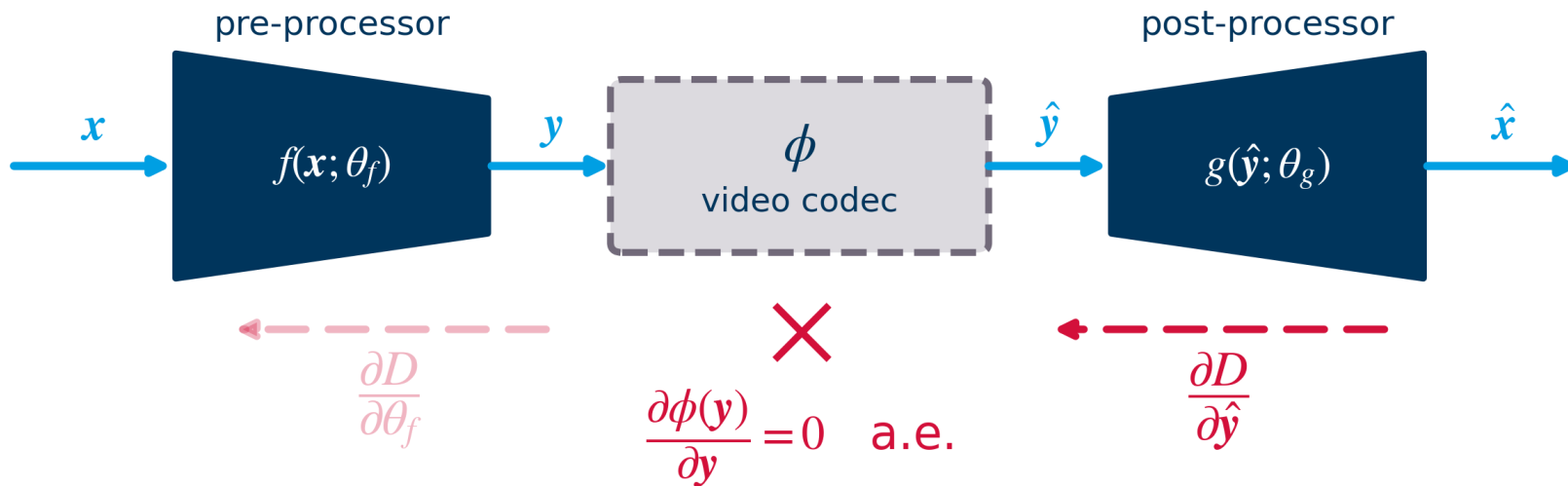
The wrapper: pre- and post-processing



The blocker: no gradient through the codec

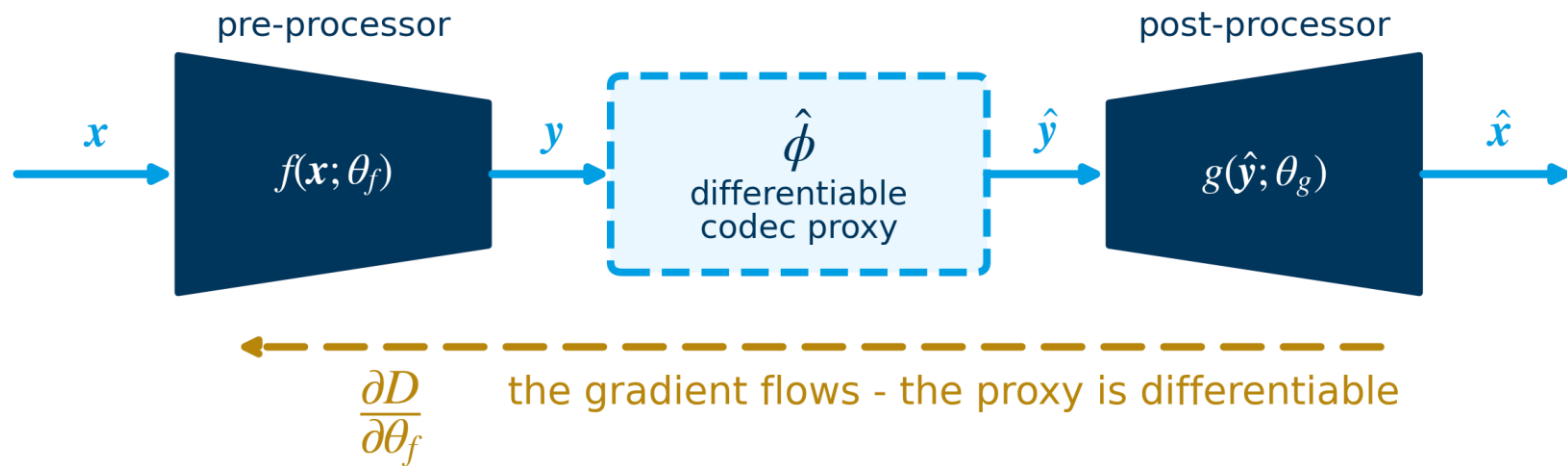
End-to-end objective

$$\theta_f^*, \theta_g^* = \arg \min_{\theta_f, \theta_g} D(x, \hat{x}) + \lambda R$$



The post-processor still trains. The pre-processor receives nothing.

Answer 1: a differentiable codec proxy



- **NN-based:** a network trained to reproduce the output of the codec
- **Hybrid-coding-based:** the coding blocks are kept, if required, replaced by a differentiable approximation (quantization, ME...)

What a proxy buys, and what it costs



What it buys

- + Differentiable by construction
- + The gradient reaches the pre-processor
- + Trains well in practice

What it costs

- **Fidelity:** these proxies may fail to capture the local artefacts introduced by the codec
- **Adaptability:** the proxy design/training may be adjusted depending on various coding parameters or standards

Answer 2: a codec surrogate gradient, not a proxy



$$\text{Forward: } \hat{y} = \phi(y) \quad \text{Backward: } \mathbf{J}_{\text{SCALED}} = \mathbf{I} - \frac{\boldsymbol{\varepsilon}(\boldsymbol{\varepsilon} - \bar{\boldsymbol{\varepsilon}})^T}{N\sigma_{\varepsilon}^2}$$

- SCALED (PCS 2025): strong results when learning a downscaler in front of the codec

Introduced as a reparameterization trick - and validated only on downscalers.

The two questions of this paper



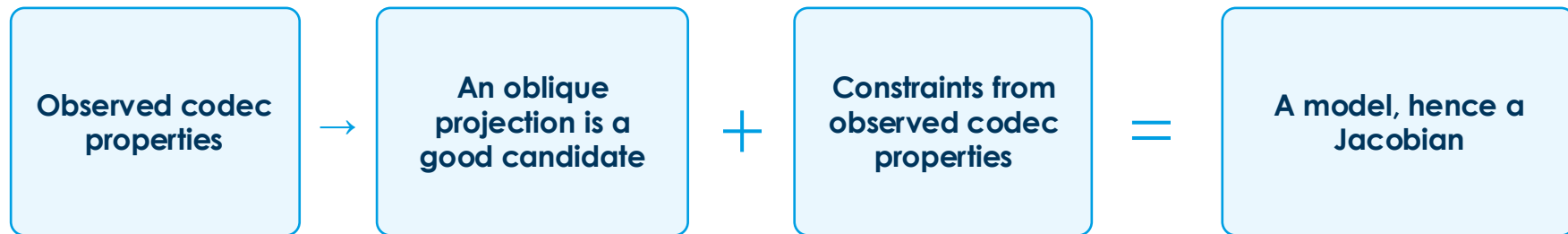
1 Can we explained why SCALED work?

2 Do the gains extend to neural wrappers?

A simple model built on the true codec statistics

The codec stays in the forward pass - the backward pass needs a model of it

$$\hat{y} = \phi(y) = y + \epsilon$$



$$\phi^2(y) \approx \phi(y) \quad \Rightarrow \quad \mathbf{P}^2 = \mathbf{P}$$

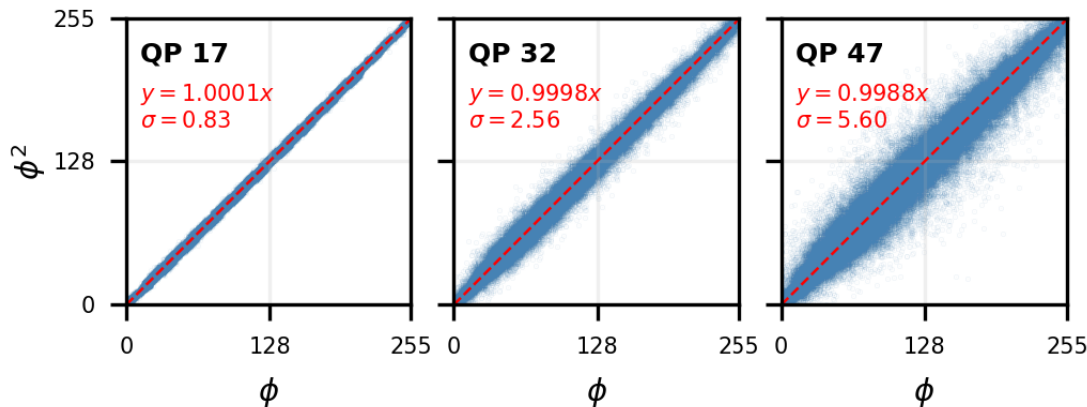


Fig. 2 - output of the codec applied twice vs once (x264, medium, 30 sequences)

Coding an already-coded image changes almost nothing: near-unity slope at every QP.

Why a projection - and what is left to choose

A codec is almost idempotent – a projection is fully idempotent

$$\hat{\phi}(y) = \mathbf{P}y = \left(\mathbf{I} - \frac{\mathbf{u}\mathbf{v}^T}{\mathbf{v}^T\mathbf{u}} \right) y$$

A linear projector, with two vectors to fix

$\mathbf{u}$ - fixed by design

The codec output is the input plus the error,
so the displacement is along $\boldsymbol{\varepsilon}$

$$\mathbf{u} = \boldsymbol{\varepsilon}$$

$\mathbf{v}$ - still to determine

Next two slides

Property 2: intensity-shift invariance



What we measure

Shifting every pixel by a constant leaves the coding error unchanged

TABLE I
INTENSITY-SHIFT INVARIANCE OF TWO CODECS (*medium* PRESET, QP 32).
VALUES REPORT $\mathbb{E}[\phi(\mathbf{x} + \delta) - \phi(\mathbf{x}) - \delta]$ AVERAGED OVER 30
SEQUENCES FROM [15]; 0 INDICATES INVARIANCE.

Codec	Translation offset δ					
	-5	-3	-1	+1	+3	+5
x264	0.015	0.002	-0.002	-0.012	-0.007	-0.027
VPenC	0.007	0.017	-0.005	-0.005	-0.011	-0.018

Table I - measured deviation < 0.03 grey level

What it constrains

The projector must commute with a global shift

$$\mathbf{P}(\mathbf{y} + \delta \mathbf{1}) = \mathbf{P}\mathbf{y} + \delta \mathbf{1}$$



$$\mathbf{v}^T \mathbf{1} = 0$$

$\mathbf{v}$ must be zero-mean

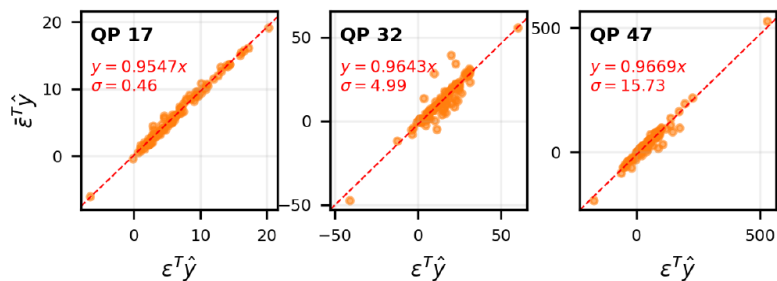
Property 3: a non-centered, orthogonal error

What we measure

The mean error is small but non-zero, and the centered error is orthogonal to the codec output

What it constrains

The candidate vector v must be orthogonal to the codec output $\hat{y}$.



$$v^T \hat{y} \approx 0$$

Fig. 3 - centered error vs codec output (x264, 100 sequences)

Finding the projector that satisfies all the conditions

v^* is the intersection of the modeling constraints : v is zero-mean, and must be orthogonal to the codec output $\hat{y}$

$$v^* \in \{v : \mathbf{1}^T v = 0\} \cap \{v : v^T \hat{y} = 0\}$$



$$v^* = \epsilon - \bar{\epsilon} \quad \Rightarrow \quad \hat{\phi}(y) = \left(\mathbf{I} - \frac{\epsilon(\epsilon - \bar{\epsilon})^T}{N\sigma_\epsilon^2} \right) y$$

Differentiating the model gives SCALED

Differentiate the model with respect to y yields:

$$\mathbf{J}_{\text{Proj}}(\mathbf{y}) = \frac{\partial \hat{\phi}(\mathbf{y})}{\partial \mathbf{y}} = \mathbf{I} - \frac{\boldsymbol{\varepsilon} (\boldsymbol{\varepsilon} - \bar{\boldsymbol{\varepsilon}})^T}{N\sigma_{\varepsilon}^2} = \mathbf{J}_{\text{SCALED}}$$

Answer to question 1: SCALED is the gradient of the MSE-optimal projector consistent with three measured codec properties

Why this gradient behaves well



Stability

$$\mathbf{P}^2 = \mathbf{P}$$

$$\lambda(\mathbf{J}_{\text{Proj}}) \in \{0, 1\}$$

$$\rho(\mathbf{J}_{\text{Proj}}) = 1$$

Bounded spectrum: the gradient cannot explode

Zero-shot

Parameter-free: the projector follows the measured error statistics, so a change of QP or of codec needs no retraining

Fidelity

The forward pass runs the true codec and the backward pass uses the real coding noise - no proxy, no domain gap

True codec forward, surrogate backward

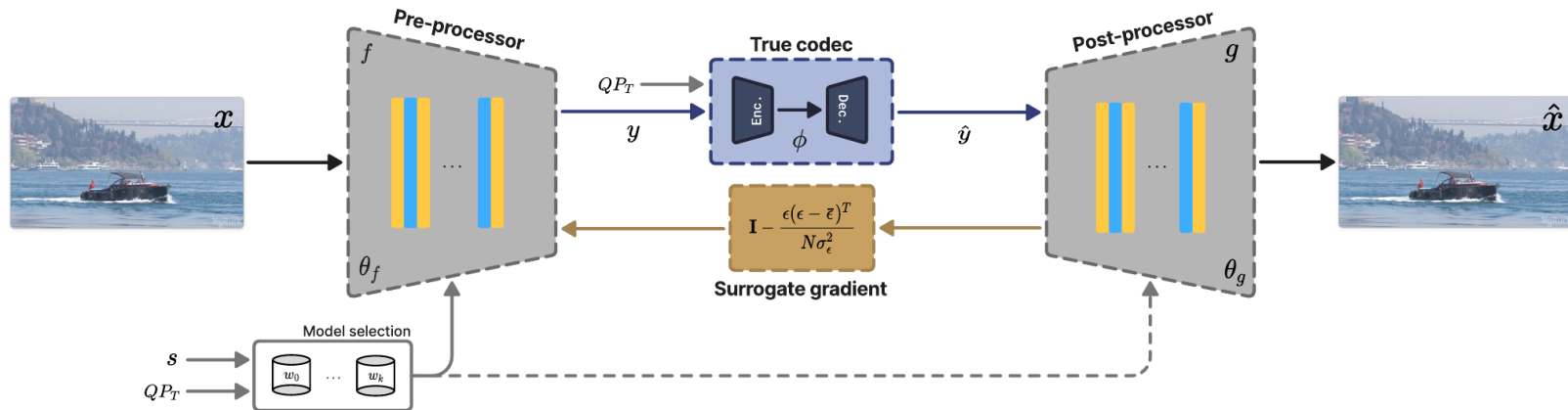


Fig. 1 - blue: forward pass through the true codec. Yellow: backward pass through the projection Jacobian.

- U-Net pre- and post-processor, residual domain
- The surrogate is rebuilt from the error of that very encode

Rate-distortion objective

$$\theta_f^*, \theta_g^* = \arg \min_{\theta_f, \theta_g} \|x - g(\hat{y}; \theta_g)\|_2^2 + \lambda \hat{\psi}(f(x; \theta_f))$$

Encoding

- x264 (H.264/AVC), medium preset
- **VVenC** (H.266/VVC), random access
- Constant QP $\in \{17, 23, \dots, 47\}$
- Scale ratios $s \in \{1, 1/2, 1/4\}$

Test set

- 7 UVG sequences + 24 Xiph sequences
- 1080p, 8-bit, 4:2:0
- One model per scale ratio and per QP

Metric

- BD-Rate on PSNR, SSIM, VMAF, VMAF-NEG
- Reference: codec alone at $s = 1$, else Lanczos $\downarrow$ / Bicubic $\uparrow$

Baseline

- Hybrid coding paradigm codec proxy from Guleryuz et al.* (2024)
- Open source code on : github.com/google/sandwiched_compression

Result 1: our surrogate vs the state of the art

BD-Rate (negative is better) - reference: the codec alone at s = 1, resampling otherwise

Ratio	Method (Forward / Backward)	x264 (H.264/AVC) [22]				VVenC (H.266/VVC) [23]			
		PSNR	SSIM	VMAF	VMAF-NEG	PSNR	SSIM	VMAF	VMAF-NEG
×1	Proxy [15] / Proxy [15]	-0.56%	-0.38%	5.21%	3.18%	1.66%	5.02%	-1.18%	-1.33%
	Proxy [15] / Proxy+Proj. quant. (Ours)	-0.21%	-11.96%	2.53%	1.30%	-0.15%	-0.06%	-0.12%	-0.11%
	True codec / Surrogate (Ours)	-7.70%	-12.45%	-4.25%	-4.01%	-1.34%	-2.14%	-1.91%	-1.65%
×1/2	Proxy [15] / Proxy [15]	-5.33%	-7.82%	4.81%	3.61%	-2.11%	2.11%	-4.16%	-4.66%
	Proxy [15] / Proxy+Proj. quant. (Ours)	-8.30%	-8.39%	-0.87%	-1.85%	-2.50%	0.39%	-0.82%	-1.87%
	True codec / Surrogate (Ours)	-15.66%	-11.78%	-11.90%	-11.68%	-8.17%	-4.76%	-5.81%	-5.72%
×1/4	Proxy [15] / Proxy [15]	-10.62%	-13.14%	-10.35%	-10.91%	-8.40%	-8.57%	-15.21%	-15.39%
	Proxy [15] / Proxy+Proj. quant. (Ours)	-15.99%	-18.97%	-8.40%	-9.28%	-9.70%	-8.16%	-9.99%	-10.95%
	True codec / Surrogate (Ours)	-23.59%	-21.01%	-36.17%	-35.97%	-20.07%	-13.47%	-30.56%	-30.55%

- In each group: **Proxy / Proxy** = state of the art [15] · **True codec / Surrogate** = ours

SOTA = the sandwiched-compression proxy, Guleryuz et al., arXiv:2402.05887. Full table (SSIM, VMAF-NEG) in the backup slides.

- **Deployable, hard to train:** the codec sits in the middle, with no usable derivative
- **Two routes:** replace the codec by a proxy, or keep it and repair the backward pass
- **Q1 – SCALED is not a heuristic:** the gradient of the MSE-optimal projector
- **Q2 - it extends to wrappers:** outperforms standard proxy at every ratios/codec standards configurations
- **One derivation, reusable:** it also replaces the STE inside any proxy

One geometric model, no extra network, no extra training - and the true codec stays in the loop.

Thank you - questions?

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[MediaKind.com](https://mediakind.com)

- **A hybrid coding paradigm, made differentiable**

Fixed 8×8 partitioning · DC intra prediction · motion estimation from optical flow, computed offline

- **Quantization: a straight-through estimator**

The forward value is quantized, the backward derivative is forced to one: $\partial Q / \partial x = 1$

- **Rate proxy: the number of surviving DCT coefficients, smoothed**

$$\|c\|_0 = \sum_i \mathbb{1}\{|c_i| \geq \Delta/2\} \approx \sum_i \log\left(1 + \frac{|c_i|}{\Delta}\right)$$

c = DCT coefficients of the codec input, Δ = quantization step; the scale is calibrated per image so that the proxy matches the true bitrate

Training

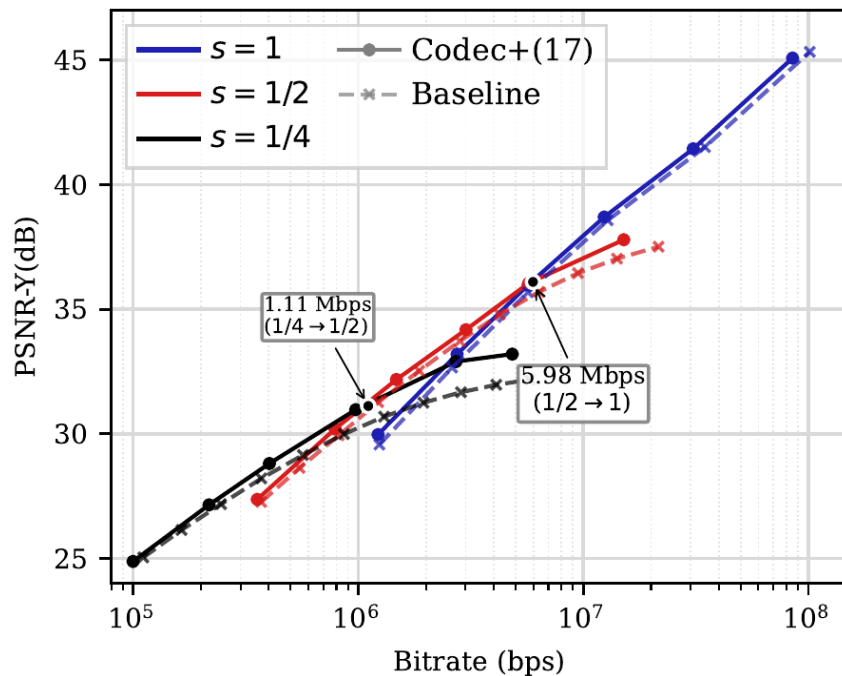
- Google sandwiched-compression dataset
- 256×256 YUV 4:4:4, 10 frames
- Chroma sub- and upsampled to simulate 4:2:0
- 500 epochs, Adam ($\eta = 10^{-4}$)
- Weighted MSE 4:1:1
- One model per scale ratio and per target QP

Networks

- U-Net pre- and post-processor
- Residual domain
- (32, 64, 128, 256) filters $\rightarrow$ 512-channel bottleneck
- Full input resolution
- Bicubic $\downarrow$ / Lanczos $\uparrow$ for scaling

Proxy

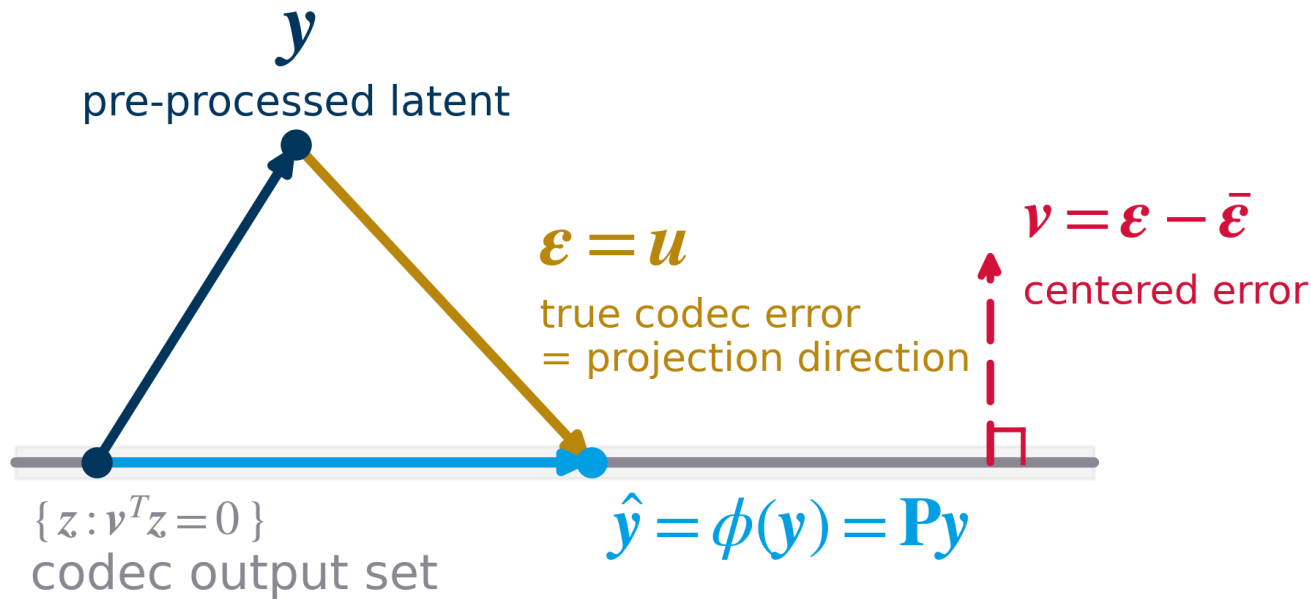
- Baseline: sandwiched-compression proxy
- Motion offline via U-Flow optical flow
- Rate proxy $\approx \ell_0$ norm of DCT coefficients
- STE replaced by the projection surrogate



- Solid: codec + our wrapper; dashed: resampling baseline
- Above the baseline at every operating point, for the three ratios
- Switching points move: 1.11 Mbps for $\times 1/4 \rightarrow \times 1/2$, 5.98 Mbps for $\times 1/2 \rightarrow \times 1$
- **Directly usable for an adaptive-bitrate ladder**

Fig. 4 - convex hull, x264 medium, average over the test set

Backup - the projection, geometrically



An oblique projection: the direction is the true codec error, the normal is its centered version.

TABLE II
COMPARISON OF BD-RATE PERFORMANCE USING x264 AND VVenC. REFERENCE IS CODEC IF $s = 1$, ELSE *Lanczos* $\downarrow$ *Bicubic* $\uparrow$.

Ratio	Method (Forward / Backward)	x264 (H.264/AVC) [22]				VVenC (H.266/VVC) [23]			
		PSNR	SSIM	VMAF	VMAF-NEG	PSNR	SSIM	VMAF	VMAF-NEG
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